SCX 后续任务

最后更新：2026-06-25（多 Agent 实现日）

Phase 1：数学框架　完成

- ☒ 核心定义 (R_m(s), SCX_m(s), V(s), L(s), D(s), N(x))
- ☒ 三个命题 + 证明
- ☒ 数据四分类规则
- ☒ SCX-Compress 冗余压缩理论
- ☒ 算法: State-Certified Active Learning

Phase 2：Python 实现　完成

- ☒ `scx/core/` — SCXConfig, SCXFramework, SCXMetrics
- ☒ `scx/state/` — StateSpace, StateDiscovery, StateAssignment, StateMetrics
- ☒ `scx/expert/` — ExpertRegistry, ExpertReliability, ExpertRouter, ExpertConflict
- ☒ `scx/valuation/` — LearnabilityScore, NoiseScore, RedundancyScore, DataClassifier, StateValue
- ☒ `scx/action/` — ActionPolicy, AcquisitionStrategy, CompressStrategy
- ☒ `scx/utils/` — helpers, DataLoader, Visualizer, Evaluation
- ☒ 测试套件 (259 tests, 全部通过)
- ☒ pyproject.toml

Phase 3：实验　待执行

- ☐ **合成实验**：运行 `experiments/synthetic/run_experiment.py`
  - ☐ 4 状态 + 3 专家 + 噪声 + 冗余数据生成
  - ☐ SCX 全流程 vs 4 baselines
  - ☐ 可视化输出 (state map, reliability heatmap, value bar)
- ☐ **通用 ML 实验**：CIFAR-100 with label noise
  - ☐ 构造 multi-expert 场景
  - ☐ 对比 LESS/random/uncertainty/diversity/SCX
- ☐ **MLIP 案例**：AlGaN residual map
  - ☐ ACE descriptor 提取
  - ☐ 状态空间构造
  - ☐ 3 expert (ACE/NEP/MACE) 可靠性估计

Phase 4：竞争分析　运行中

- ☐ 竞争分析 1: 数据估值 + 主动学习（agent 运行中）
- ☐ 竞争分析 2: MoE + 蒸馏 + 专家合并（agent 运行中）
- ☐ 综合竞争格局文档

Phase 5：论文写作

- ☐ Section 1: Introduction
- ☐ Section 2: Problem Formulation
- ☐ Section 3: State-Conditioned eXpertise
- ☐ Section 4: State-Conditioned Data Value
- ☐ Section 5: SCX Policy / Algorithm
- ☐ Section 6: Synthetic Experiments

- ☐ Section 7: General ML Experiments
- ☐ Section 8: Scientific ML Case Study
- ☐ Section 9: Discussion

Phase 6：投稿

- ☐ arXiv preprint
- ☐ Workshop 投稿
- ☐ 正式投稿 (TMLR / AISTATS / NMI)